

FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

A board each, five tasks, everything pre-wired

- Upload, change two numbers, upload again
- You wire nothing today

Actuators are on the benches around you

- Go and feel the ones your task does not use

Finish by building one signal

- Someone else has to read it

RUN SHEET

Times are offsets from the start. The only hard checkpoint is 0:25: everyone has task 01 running.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	15 min	Boards out, task 01 running for everyone
0:25	60 min	Tasks 02 to 05, at your own pace
1:25	10 min	Round-the-room: what surprised you
1:35	20 min	Blind test on task 05
1:55	5 min	Pack down

FIVE TASKS

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

- 01 Make something spin coin motor**
- 02 Make something tap solenoid**
- 03 Make something warm Peltier tile**
- 04 Make it notice you a wire and foil**
- 05 Make it answer back both together**

1 to 3 are output. 4 is input. 5 is both.

Nobody is expected to finish all five.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments, roughly in order of difficulty
- The last one is usually the interesting one



github.com/gust1527/multimodal-actuator-workshop

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MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT



THE ONE THAT SURPRISES PEOPLE

Touch input for the price of a wire. The thinking is where it costs you.

reading > baselineAverage() * 1.01

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table with a sign. Wander over between tasks.

- Capacitive touch. A wire. That is task 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are

Blind test: they name all three



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- Note which two got confused, and why

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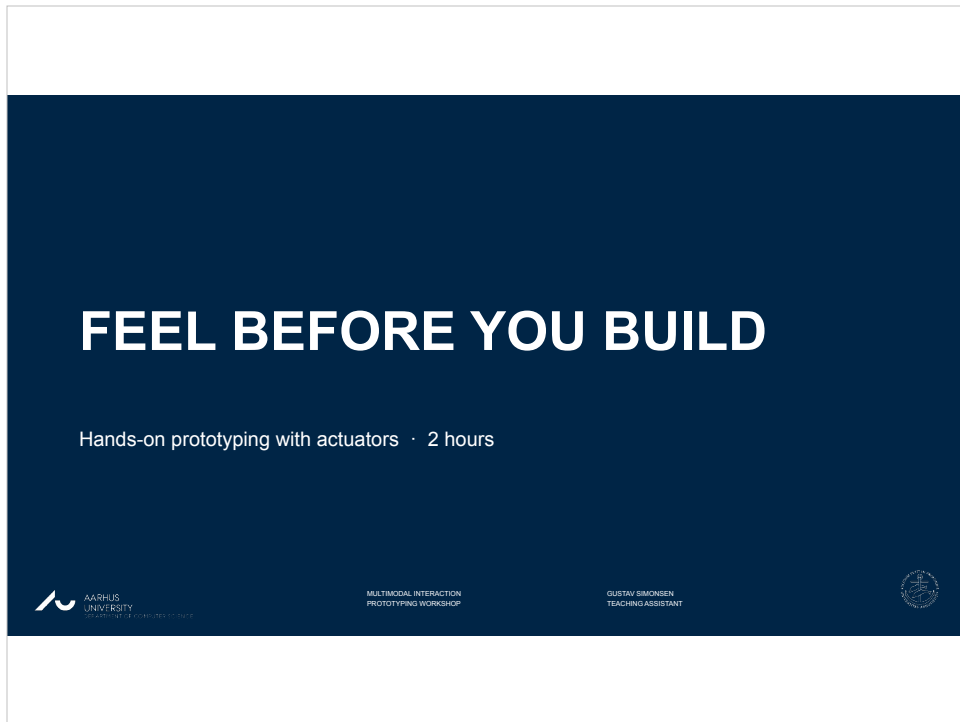
THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,
you built a visual interface with a motor glued to it.

GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop



0:00. Start here, and do not read the slide.

Open with the demo, not with talking. Have the coin motor and the solenoid both wired and running before anyone sits down.

Buzz a phone-style notification on the motor. Then send the same message on the solenoid. Ask which felt more urgent. Let them answer. Nobody needs theory for that, and it is the argument for the whole session.

WHY WE ARE HERE

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- Not because touch was wrong
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Today: decide it feels cheap, pick something else

- Better now than in week 46



Say the full version out loud, the slide only carries the skeleton:

"You can read a datasheet. You have never held one of these. Today you put a vibration motor against your own wrist, decide it feels cheap, and pick something else. Better now than in week 46 with the demo three days out."

Every year projects end up on the screen and the speaker, and touch gets designed on paper then dropped. Two minutes, then move.

HOW TODAY WORKS

A board each, five tasks, everything pre-wired

- Upload, change two numbers, upload again
- You wire nothing today

Actuators are on the benches around you

- Go and feel the ones your task does not use

Finish by building one signal

- Someone else has to read it



Set the shape clearly, because it is not a lecture and they will wait to be lectured at.

A board each. Five tasks. Everything pre-wired. They upload, change numbers, upload again.

The actuators they are NOT using sit on the benches around the room. Tell them explicitly to go and feel those between tasks. That is the comparison that used to be a guided tour; it now happens on their own initiative, which works better anyway.

RUN SHEET

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Show it, do not narrate it. Fifteen seconds.

The one hard checkpoint is 0:25. Everyone should have task 01 running by then.

If someone is still fighting drivers at 0:30, pair them with a group that is working rather than debugging it alone.



0:10. Boards out.

Get EVERYONE through task 01 before anyone races ahead. It is the shortest and it flushes out the setup problems: wrong board, wrong port, missing driver.

Once task 01 runs for a pair, they can move at their own pace.

FIVE TASKS, ONE BOARD EACH

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- | | | |
|----|---------------------|-----------------|
| 01 | Make something spin | coin motor |
| 02 | Make something tap | solenoid |
| 03 | Make something warm | Peltier tile |
| 04 | Make it notice you | a wire and foil |
| 05 | Make it answer back | both together |

1 to 3 are output. 4 is input. 5 is both.

Nobody is expected to finish all five.



Ninety seconds on this slide, no more.

Name the arc, it is the point: 1 to 3 are output, 4 is input, 5 is both. By task 5 they have built a complete interaction loop, which is the shape of most project work.

Say clearly that nobody is expected to finish all five. Getting 01 working and then properly playing with 02 beats rushing all of them.

Task 05's last experiment IS the closing brief, so the work carries over.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

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MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT



Say the format once here rather than at every bench.

CHANGE ME at the top: two or three numbers, already set to something that works. THINGS TO TRY at the bottom: four experiments, easy to hard, and the last one usually asks for a rewrite rather than a tweak.

Upload problems, in the order they will hit you:

- * wrong board or port under Tools
- * ADCTouch not installed (tasks 04 and 05)
- * solenoid task: board resets on fire means the bench supply is off

THE ONE THAT SURPRISES PEOPLE

Touch input for the price of a wire. The thinking is where it costs you.

`reading > baselineAverage() * 1.01`

Threshold is relative, never absolute

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Baseline only learns while untouched

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Slow down here. This is the single idea most worth stealing from today, and it is worth interrupting the room for once most people have reached task 04.

Two things students get wrong:

1. A fixed threshold. Works in the morning, fails after lunch, because the reading drifts with humidity and mains hum.

2. Feeding the baseline buffer always. A long press then teaches the baseline that a finger is normal, and the touch silently stops working. The buffer must only learn while untouched

INPUTS, AND PRIOR ART



1:25. Pull the room back together.

One sentence per group: what surprised you.
No slides, no laptops. Ten minutes.

Expect someone to report the Peltier delay. Let
that land, it is far more
persuasive from a peer than from you.

INPUTS WORTH KNOWING ABOUT

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Mention, do not teach. These sit on one table with a sign.

The honest warnings matter more than the specs: PPG is useless once the hand moves, GSR drifts so only relative change works, FSRs need per-pad calibration.

Push the phone option hard. For a lot of projects it beats everything else on the table and needs no soldering at all.

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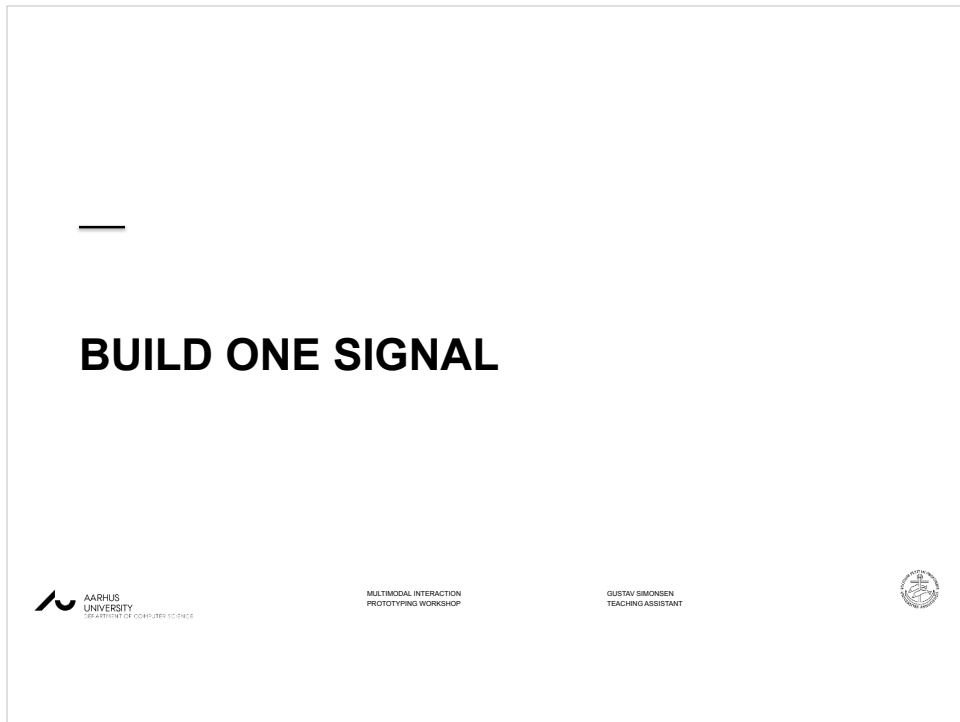


Five minutes, and put the actual repos on screen if you can.

The argument: this was built by students at your stage, not by a lab with a budget. The gap between "we should use touch" and "it moves when you touch it" is smaller than it looks.

The moving screen has a browser control panel over the serial port, which is what you will wish you had at the project demonstrations.

Note for you: the sock repo has a hardcoded Wi-Fi password and live patient IDs



1:35. The blind test.

Most groups will already have something from task 05. This is where it gets tested by someone who did not build it.

THE BRIEF

30 minutes. Not long enough to be precious about it.

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Blind test: they name all three

- Note which two got confused, and why

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Read the brief out once, then get out of the way. Full wording, since the slide is only the bones:

"Pick one actuator. Encode three messages: arrived, something wrong, finished. Vary only the rhythm of the pulses and how strong they are. Then hand it to another group with the screen turned away and see if they can name all three."

If someone asks to add a second actuator, say no. The constraint is the exercise.

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,
you built a visual interface with a motor glued to it.



The line worth repeating twice.

Run the test: swap groups, receiver looks away, guess all three.

Then have each group name which two got confused and what would have separated them. That confusion is the actual learning, not the successful ones.



Point them at the repo. Every task sketch is commented for them, and the bench test sketches are there too if they want to see the minimal version.

Last thing to say: pick your actuator before you design the interaction, not after. Today was so they can do that from experience rather than a datasheet.

Pack down: everything back in the box, including the resistors. It will not all come back.

If anyone wants to borrow parts for the project weeks, now is when they ask